Sensitivity, especially Omitted Variable Bias

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Old idea. Cornfield et al. (1959) on smoking and lung cancer:

> *. . . an unobserved confounder . . . would need to lead to a nine-fold increase in the odds of smoking to explain away the association . . . and it was asserted that such a strong unobserved confounder was very unlikely to exist.*

Effects of violence on attitudes: angry or weary?

Among Darfurian refugees in eastern Chad, did exposure to violence on average make people more willing to make concessions to achieve peace ("weary"), or more desiring of retribution ("angry")?

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Strategy: SOO/CI conditioning on village and gender.

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4. How can we make informed guesses about plausible confounding?

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We focus on OVB-based: scale-free in R^2, handles multiple/non-linear confounders, plugs into the regression workflow we just built.

A short detour: Frisch–Waugh–Lovell

Suppose we run OLS of Y on D and $\boldsymbol{X}$:

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In words: *multivariate OLS reduces to univariate OLS, after partialling out* $\boldsymbol{X}$,

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This is the central tool for everything that follows.

Omitted variable bias: setup

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Whether the "full" regression is wise, or its coefficient is the quantity of interest, is a separate question — here we just compute the consequence of omission.

OVB via FWL: bias $= \hat{\gamma}\hat{\delta}$

Restating FWL,

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$$\boxed{\widehat{\text{bias}}(\hat{\tau}_{\text{res}}) \ = \ \hat{\gamma}\,\hat{\delta} \ = \ \text{impact} \times \text{imbalance}}$$

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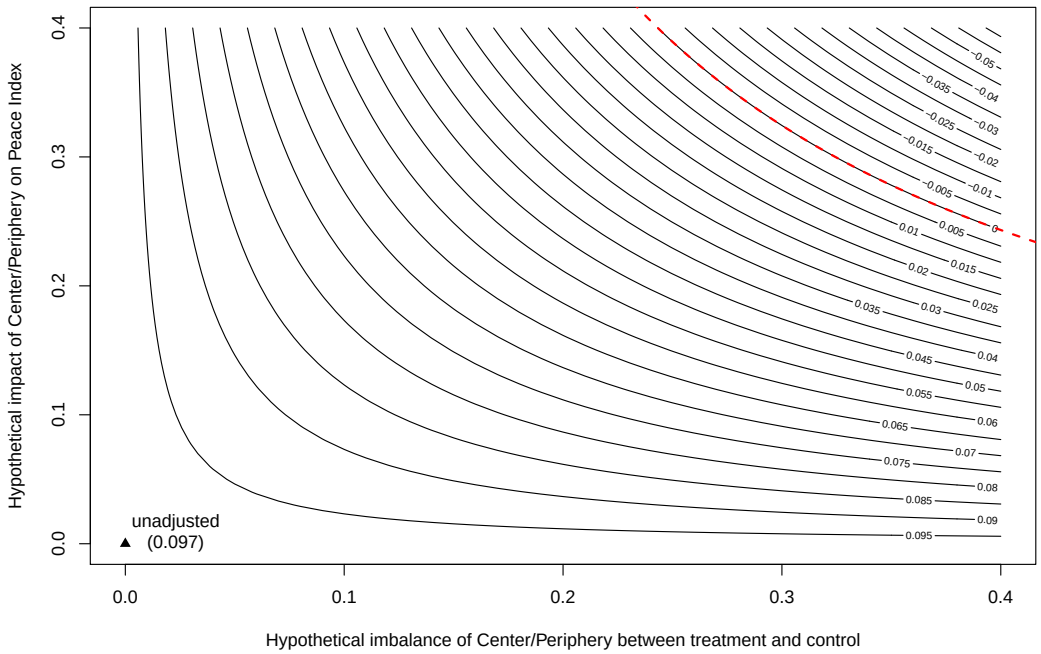

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- Scale-free, intuitive, bounded in $[0, 1]$
- With this we will derive bias *and* variance, do bounds, handle all confounders jointly, and more.

What is the partial R^2?

The partial correlation is simply correlation *after* partialling out $\boldsymbol{X}$:

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Or: "fraction of D's explained by Z beyond what $\boldsymbol{X}$ explained":

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Direction-symmetric: $R^2_{D \sim Z|\boldsymbol{X}} = R^2_{Z \sim D|\boldsymbol{X}}$

Deriving bias in the R^2 parameterization

By FWL,

$$\begin{aligned}\widehat{\text{bias}} &= \hat{\gamma}\hat{\delta} \\ &= \left(\frac{\text{cov}(Y^{\perp \boldsymbol{X},D}, Z^{\perp \boldsymbol{X},D})}{\text{var}(Z^{\perp \boldsymbol{X},D})}\right)\left(\frac{\text{cov}(D^{\perp \boldsymbol{X}}, Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})}\right) \\ &= \text{cor}(Y^{\perp \boldsymbol{X},D}, Z^{\perp \boldsymbol{X},D}) \cdot \text{cor}(D^{\perp \boldsymbol{X}}, Z^{\perp \boldsymbol{X}}) \\ &\quad \cdot \frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})} \cdot \frac{\text{sd}(Z^{\perp \boldsymbol{X}})}{\text{sd}(Z^{\perp \boldsymbol{X},D})}.\end{aligned}$$

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$$\boxed{|\widehat{\text{bias}}| = \sqrt{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}}\, R^2_{D\sim Z|\boldsymbol{X}}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}} \cdot \frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})}.}$$

Bias anatomy

An OLS variance fact (for now and later):

$$\widehat{\text{se}}(\hat{\tau}_{\text{res}}) \;=\; \frac{\text{sd}(Y^{\perp \boldsymbol{X}, D})}{\text{sd}(D^{\perp \boldsymbol{X}})\,\sqrt{\text{df}}}.$$

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$$\widehat{\text{se}}(\hat{\tau}_{\text{res}}) \;=\; \frac{\text{sd}(Y^{\perp \boldsymbol{X}, D})}{\text{sd}(D^{\perp \boldsymbol{X}})\sqrt{\text{df}}}.$$

Nice to rewrite bias then as:

$$|\widehat{\text{bias}}| \;=\; \sqrt{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}}\, R^2_{D\sim Z|\boldsymbol{X}}}{1-R^2_{D\sim Z|\boldsymbol{X}}}} \cdot \frac{\text{sd}(Y^{\perp \boldsymbol{X}, D})}{\text{sd}(D^{\perp \boldsymbol{X}})} \;=\; \widehat{\text{se}}(\hat{\tau}_{\text{res}}) \cdot \sqrt{\underbrace{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}}\, R^2_{D\sim Z|\boldsymbol{X}}}{1-R^2_{D\sim Z|\boldsymbol{X}}}}_{\text{BF}^2} \cdot \text{df}}.$$

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Consider the "relative bias", i.e. the proportion of $\hat{\tau}_{res}$ that is bias. Substitute $|\hat{\tau}_{\text{res}}| = |t_D|\cdot\widehat{\text{se}}(\hat{\tau}_{\text{res}})$ in the denominator ($|t_D|$ = absolute t-stat for D):

$$\frac{|\widehat{\text{bias}}|}{|\hat{\tau}_{\text{res}}|} \;=\; \frac{\widehat{\text{se}}(\hat{\tau}_{\text{res}})\cdot\sqrt{\text{BF}^2\cdot\text{df}}}{|t_D|\cdot\widehat{\text{se}}(\hat{\tau}_{\text{res}})} \;=\; \frac{\sqrt{\text{df}}\cdot\text{BF}}{|t_D|}.$$

Bias anatomy:

One last touch:

$$\frac{\sqrt{\text{df}} \cdot \text{BF}}{|t_D|} = \frac{BF}{|f_{Y \sim D|\boldsymbol{X}}|}$$

where $f_{Y \sim D|\boldsymbol{X}} = R_{Y \sim D|\boldsymbol{X}} / \sqrt{1 - R_{Y \sim D|\boldsymbol{X}}}$ is the (partial) Cohen's f.

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Key insights

- Numerator: BF = essential expression of confounding strength.
- Denominator: $f_{Y \sim D|\boldsymbol{X}}$ is essentially how much Y is explained uniquely by D and is *all* you need from the data to understand sensitivity. It is what confounding is competing to explain.
- And $f_{Y \sim D|\boldsymbol{X}}$ is not function of sample size: more data doesn't help you fight confounding.

Variance anatomy

Same OLS variance fact, applied to the *full* regression:

$$\widehat{\text{se}}(\hat{\tau}) \;=\; \frac{\text{sd}(Y^{\perp \boldsymbol{X}, D, Z})}{\text{sd}(D^{\perp \boldsymbol{X}, Z})\sqrt{\text{df} - 1}}.$$

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Two ratios of variances are partial R^2's:

$$\frac{\text{sd}^2(Y^{\perp \boldsymbol{X},D,Z})}{\text{sd}^2(Y^{\perp \boldsymbol{X},D})} = 1 - R^2_{Y\sim Z|\boldsymbol{X},D}, \quad \frac{\text{sd}^2(D^{\perp \boldsymbol{X},Z})}{\text{sd}^2(D^{\perp \boldsymbol{X}})} = 1 - R^2_{D\sim Z|\boldsymbol{X}}.$$

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Combine, factoring out $\widehat{\text{se}}(\hat{\tau}_{\text{res}})$:

$$\widehat{\text{se}}(\hat{\tau}) \;=\; \widehat{\text{se}}(\hat{\tau}_{\text{res}}) \cdot \sqrt{\underbrace{(1 - R^2_{Y\sim Z|\boldsymbol{X},D})}_{\text{VRF}} \underbrace{\frac{1}{1-R^2_{D\sim Z|\boldsymbol{X}}}}_{\text{VIF}} \underbrace{\frac{\text{df}}{\text{df}-1}}_{\text{df adj.}}}.$$

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- **VRF**: confounder shrinks residual Y variance.
- **VIF**: confounder's correlation with D inflates the SE.

Two simple statistics for routine reporting

Robustness Value (RV): the level of (equal) association with D and Y that would eliminate the effect.

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Both come from t_D and df alone — you can compute them from a published regression table.

Augmented regression table: Darfur

Outcome: *Peace Index*

Treatment:	Est.	SE	t	$R^2_{Y\sim D\mid \boldsymbol{X}}$	RV	$\text{RV}_{\alpha=0.05}$
Directly Harmed	0.097	0.023	4.18	2.2%	13.9%	7.6%
df $= 783$		*Bound (1× female):* $R^2_y = 12\%$, $R^2_d = 1\%$				

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Cold-call check:

- What does this RV mean?
- What does $\text{RV}_{\alpha=0.05}$ mean?
- What does $R^2_{Y\sim D\mid \boldsymbol{X}}$ mean?

Bounding by comparison to observed covariates

"Benchmarking" interpretation: For an observed covariate X_j, define

$$k_D := \frac{R^2_{D \sim Z | \boldsymbol{X}_{-j}}}{R^2_{D \sim X_j | \boldsymbol{X}_{-j}}}, \qquad k_Y := \frac{R^2_{Y \sim Z | \boldsymbol{X}_{-j}, D}}{R^2_{Y \sim X_j | \boldsymbol{X}_{-j}, D}}.$$

"Z is at most k_D times as predictive of D, k_Y times as predictive of Y, as X_j is."

[1] This is more complicated than it sounds: the observed R^2 of observed X are themselves biased, including by collider bias from the inclusion of D. But you can construct bounds; see Cinelli & Hazlett 2020.

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This shows the implications of an assumption — you have to argue the assumption itself is believable.

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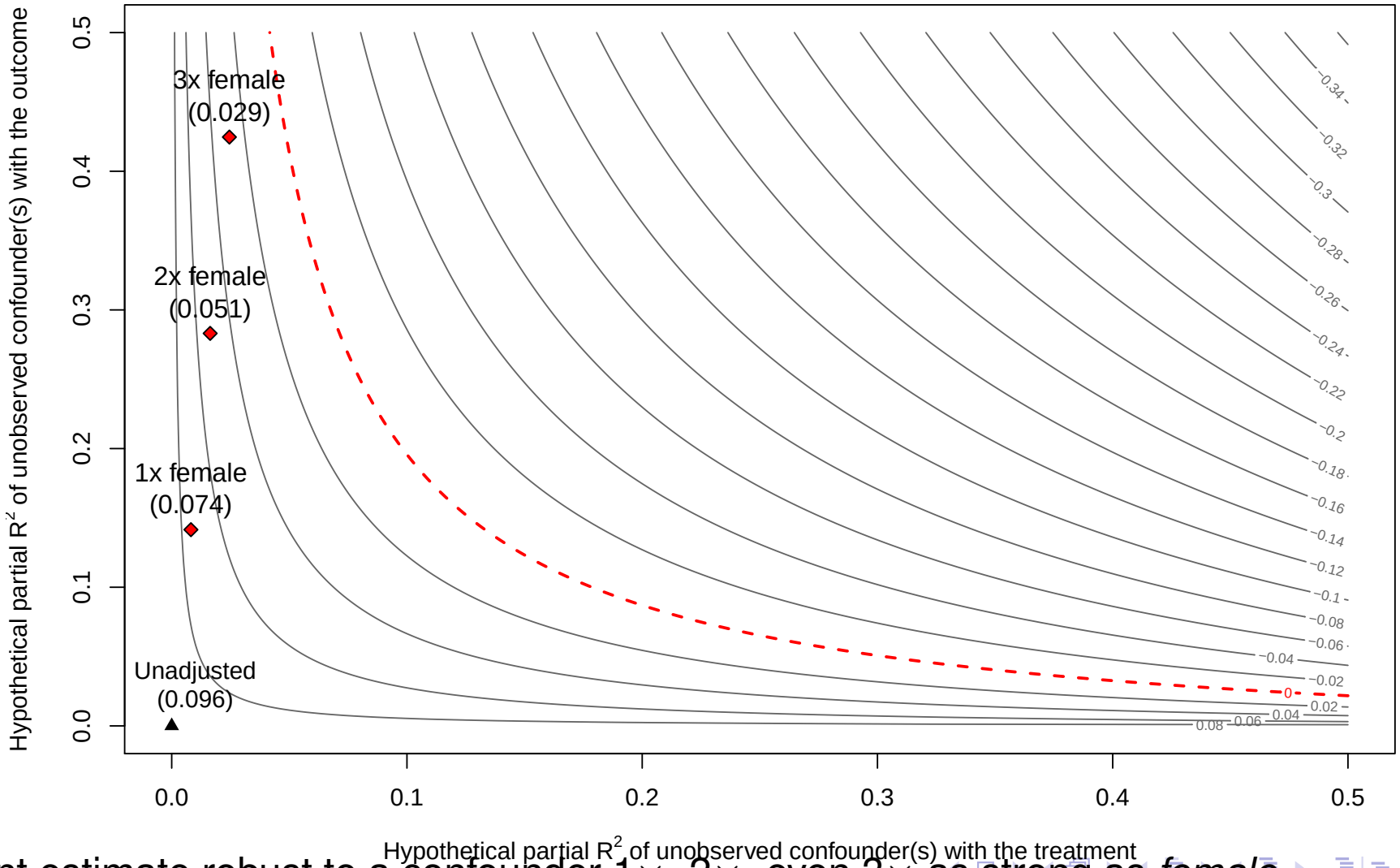
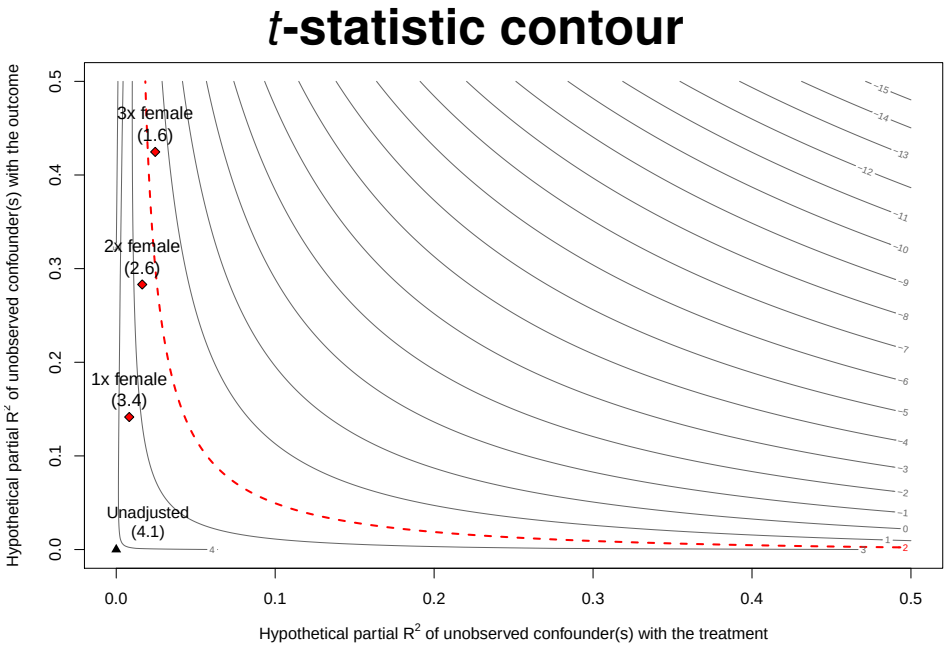
R^2 contour with bounds: Darfur

Within village, it is hard to argue *any* basis for targeting could matter more than *female*. So $X_j = female$, with various k.



Point estimate robust to a confounder 1×, 2×, even 3× as strong as *female*.

Inference and extreme scenarios



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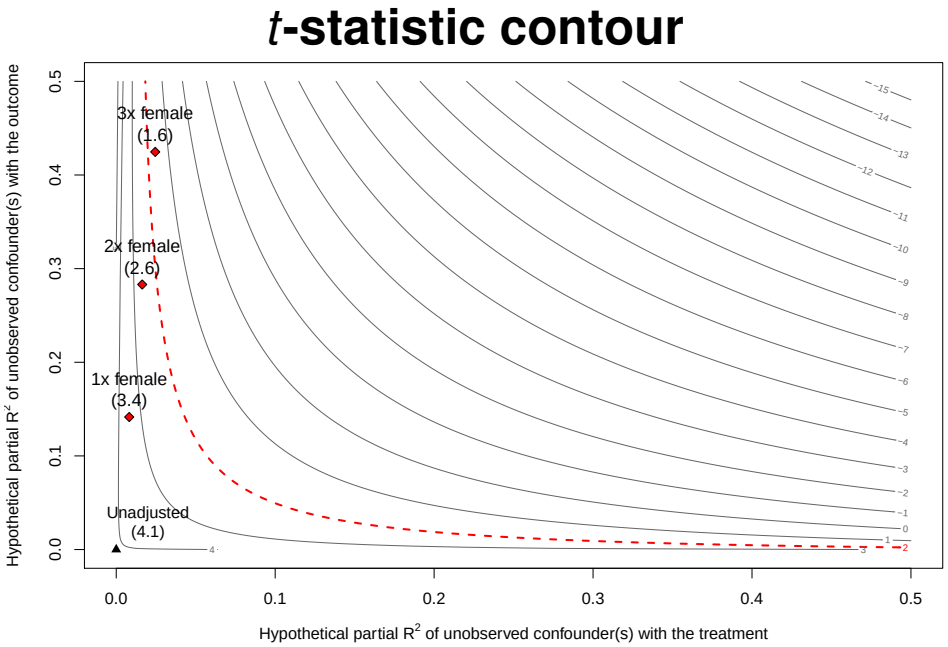

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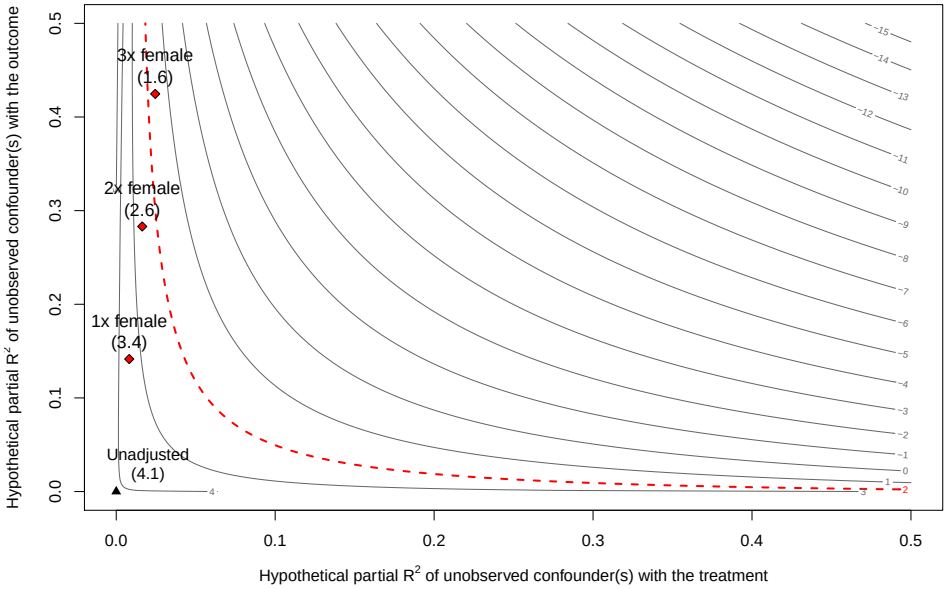
t-statistic contour



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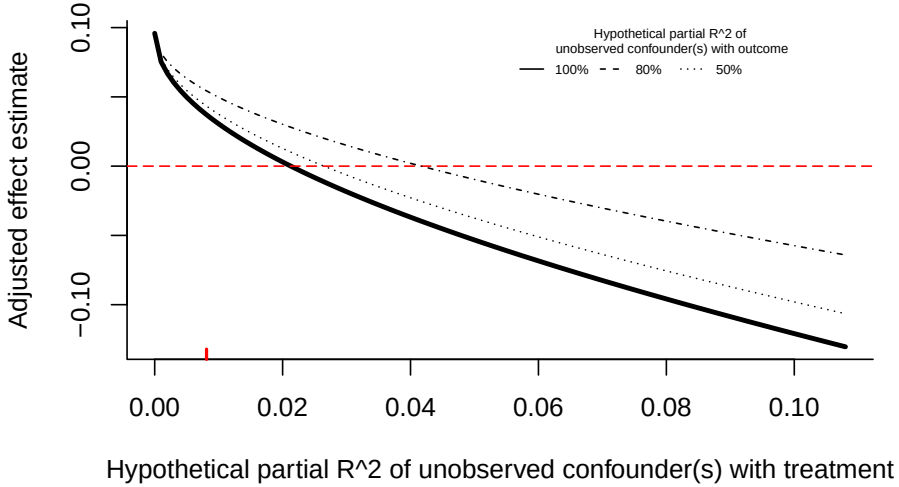

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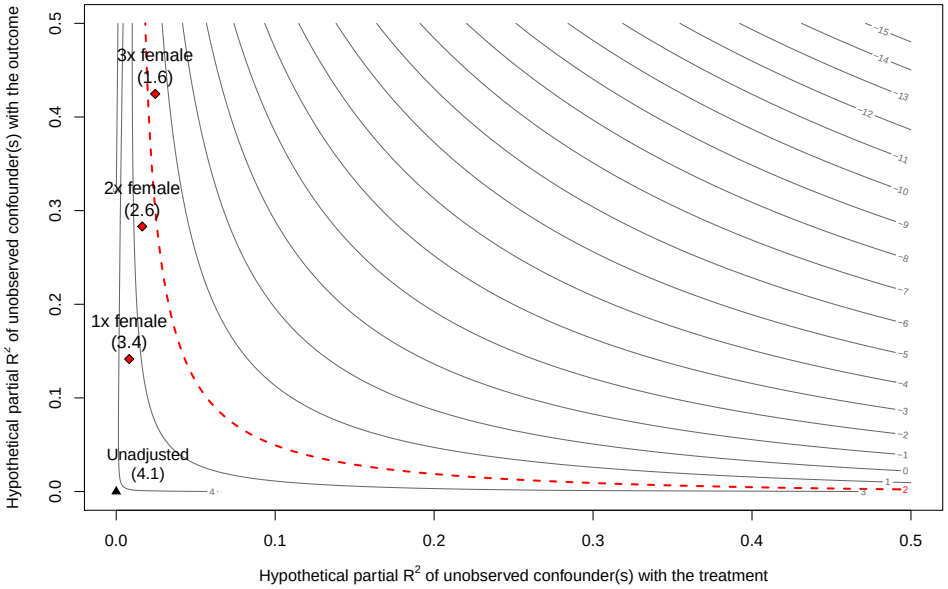

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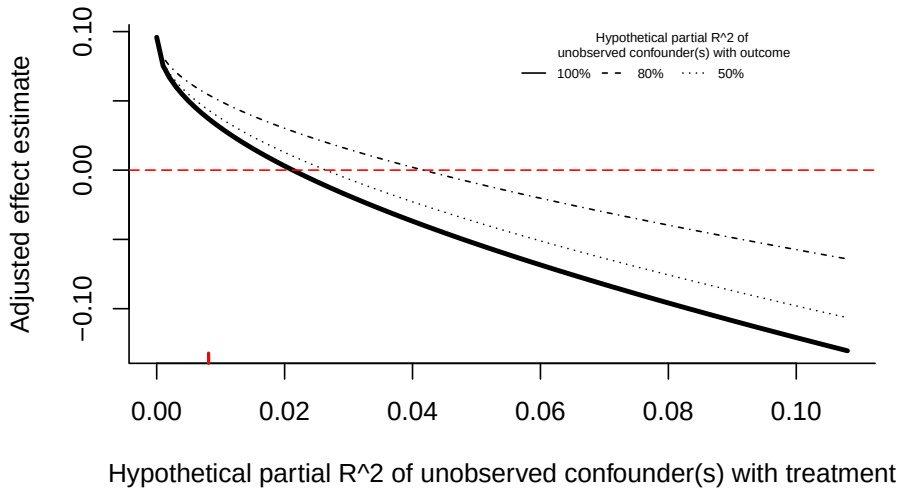

Assume $R^2_{Y\sim Z|\boldsymbol{X},D} = 1$. Then Z would still need to explain $\geq 2.2\%$ of D to zero the effect. Pay a real price when you can't bound.

How to argue it, and what readers should demand

Occasionally you might be able to argue in support of a claim, e.g. in Darfur:

> *"...aircraft dropped makeshift and unguided bombs over villages, and militia raided without concern for who they would attack — the only known major exception, due to sexual assaults, was targeting gender, which is also one of the most visually apparent characteristics.* ***A confounder twice as strong as female would be surprising.****"*

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More broadly, remember:

- No magic RV threshold: refine the debate, not hide behind a number.
- This is never "how much feasible confounding is there" — it's "how much confounding would it take to overturn the conclusion"
- The result might be informative or might show that we can't say much, but either way it is the honest conversation.

Resources

Papers (all linked from chadhazlett.com/publications):

- Cinelli & Hazlett (2020). *Making sense of sensitivity: Extending omitted variable bias.* JRSS B. — the framework
- Hazlett & Parente (2023). *From "Is it unconfounded?" to "How much confounding would it take?"* JoP. — applied / how-to-argue companion
- Cinelli, Ferwerda & Hazlett (2024). *sensemakr: Sensitivity Analysis Tools for OLS in R and Stata.* Observational Studies. — software paper
- Cinelli & Hazlett (2025). *An OVB framework for sensitivity analysis of IV.* Biometrika. — IV extension (cf. extras)

Software & tutorials:

- `sensemakr` R/Stata + worked vignettes: carloscinelli.com/sensemakr
- Carlos Cinelli's site (talks, tutorials, videos): carloscinelli.com
- Software list on my site: chadhazlett.com/software

Rosenbaum's Γ for matched pairs

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Rosenbaum's Γ for matched pairs

- A different sensitivity tradition (Rosenbaum 2002), with a single parameter $\Gamma \geq 1$ quantifying departure from zero confounding.
- Consider matched pairs i and j with $X_i = X_j$. Under SOO, both have the same $\pi(X)$.
- A binary unobserved U may make their odds of treatment differ:

$$\frac{1}{\Gamma} \leq \frac{\pi_i(1-\pi_j)}{(1-\pi_i)\pi_j} \leq \Gamma.$$

Formal sensitivity tests (Γ)

- $\Gamma = 1$: no hidden bias. $\Gamma = 2$: unit i has twice the odds of treatment as j despite identical X.

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- R: `psens()`, `hlsens()` in `rbounds`.

Rosenbaum-style sensitivity: smoking example

TABLE 4.1. Sensitivity Analysis for Hammond's Study of Smoking and Lung Cancer: Range of Significance Levels for Hidden Biases of Various Magnitudes.

Γ	Minimum	Maximum
1	< 0.0001	< 0.0001
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If a hidden confounder gave the unit with the worse outcome 6× higher odds of smoking, the adjusted estimate would lose significance.

Imbens (2003): parametric approach

- Highly parametric:
  - Assume $Y_1, Y_0 \perp\!\!\!\perp D \mid X, U$ with $U \sim \mathrm{Bernoulli}(0.5)$, $U \perp X$.
  - Logistic propensity: $P(D = 1 \mid X, U) = \dfrac{\exp(X\theta + \gamma U)}{1 + \exp(X\theta + \gamma U)}$. Here γ governs U–D strength.
  - Outcome conditionally normal: $Y \mid X, U \sim N(\alpha D + X\beta + \delta U,\ \sigma^2)$. δ governs U–Y strength.
  - Choose (γ, δ), MLE for $\hat{\alpha}(\gamma, \delta)$.
- Reparameterize (γ, δ) as partial R^2's with D and with Y for interpretability.

Trades stronger assumptions for more direct probabilistic interpretation.
Carnegie–Dorie–Hill (2016) extends with BART for $\mu_0(X)$.

OVB with instrumental variables

We can apply the same OVB machinery to IV. Walks through, applied to the Card (1993) returns-to-education example.

Running example: returns to education (Card 1993)

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- Confounding may be severe: ability, motivation, family/social.
- Card uses *proximity to college* as instrument.

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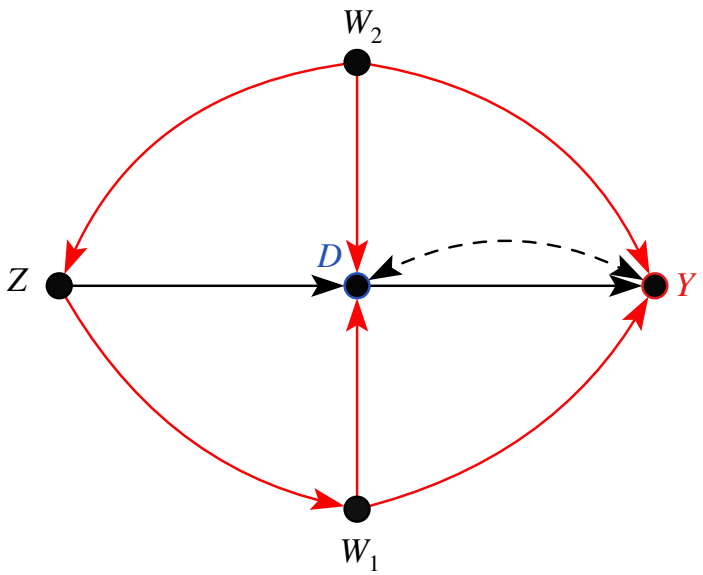

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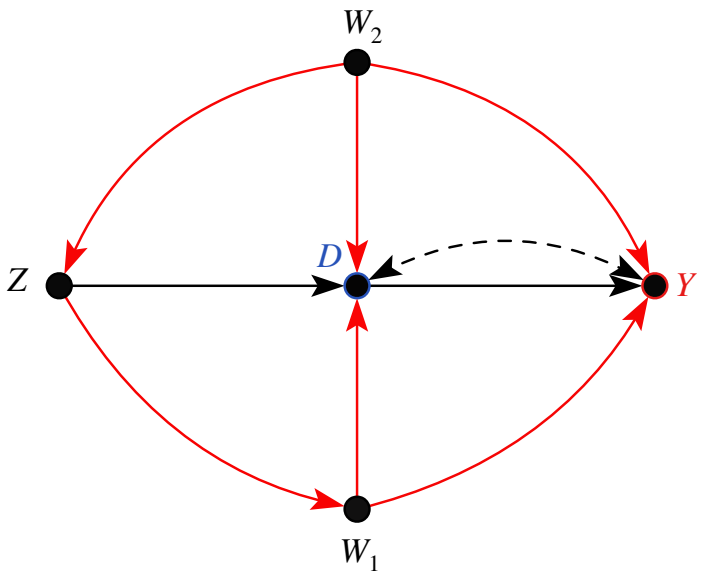

How "bad" would such violations need to be to change the conclusion?

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Same answer:

$$\hat{\tau}_{\text{2SLS}} = \hat{\tau}_{\text{ILS}} := \frac{\hat{\lambda}_{\text{res}}}{\hat{\theta}_{\text{res}}} = \frac{0.042}{0.319} \approx 0.132.$$

IV regression: Anderson–Rubin

3. Anderson–Rubin (Fieller):

$$\text{Earnings} - \tau_0 \text{Education} = \hat{\phi}_{\tau_0,\text{res}} \text{Proximity} + \boldsymbol{X}\hat{\beta}_{\tau_0,\text{res}} + \hat{\epsilon}_{\tau_0,\text{res}}$$

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How would these change had we included W?

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How would including W change inference about λ?

Same OVB analysis, different variable names

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|\widehat{\text{bias}}| &= \sqrt{\frac{R^2_{Y \sim W|Z,\boldsymbol{X}}\, R^2_{Z \sim W|\boldsymbol{X}}}{1 - R^2_{Z \sim W|\boldsymbol{X}}}\ \text{df}} \cdot \widehat{\text{se}}(\hat{\lambda}_{\text{res}}) \\
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Using the RF coefficient:

- $\text{RV}_{q,\alpha}(\lambda)$: minimal strength of association with both Z and Y needed.
- $\text{XRV}_{q,\alpha}(\lambda)$: minimal strength with Z alone, even if W explains *all* residual outcome variance.

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AR is consistent with this logic.

Sensitivity with the AR regression

$$Y_{\tau_0} = \hat{\phi}_{\tau_0,\text{res}} Z + \boldsymbol{X}\hat{\beta}_{\tau_0,\text{res}} + \hat{\epsilon}_{\tau_0,\text{res}}$$

The AR confidence interval:

$$\text{CI}_{1-\alpha}(\tau) = \left\{\tau_0 : \; t^2_{\hat{\phi}_{\text{res},\tau_0}} \leq (t^*_{\alpha,\text{df}-1})^2\right\}.$$

Replace $t^*_{\alpha,\text{df}-1}$ with $t^{\dagger\ \max}_{\alpha,\text{df}-1,\boldsymbol{R}^2}$:

$$\text{CI}^{\max}_{1-\alpha,\boldsymbol{R}^2}(\tau) = \left\{\tau_0 : \; t^2_{\hat{\phi}_{\text{res},\tau_0}} \leq (t^{\dagger\ \max}_{\alpha,\text{df}-1,\boldsymbol{R}^2})^2\right\}.$$

Analytical solution via quadratic inequality. Yields $\text{RV}_{q,\alpha}(\tau)$, $\text{XRV}_{q,\alpha}(\tau)$ and $\text{RV}_{\geq q,\alpha}(\tau)$, $\text{XRV}_{\geq q,\alpha}(\tau)$.

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Outcome: *Earnings* (log)

Instrument	Est.	SE	t	$R^2_{Y\sim Z\mid \boldsymbol{X}}$	$\text{XRV}_{q^*,\alpha}$	$\text{RV}_{q^*,\alpha}$
Proximity	0.042	0.018	2.33	0.18%	0.05%	0.67%

Bound (1× smsa): $R^2_{Y\sim W\mid Z,\boldsymbol{X}} = 2\%$, $R^2_{Z\sim W\mid \boldsymbol{X}} = 0.6\%$, $t^\dagger = 2.55$
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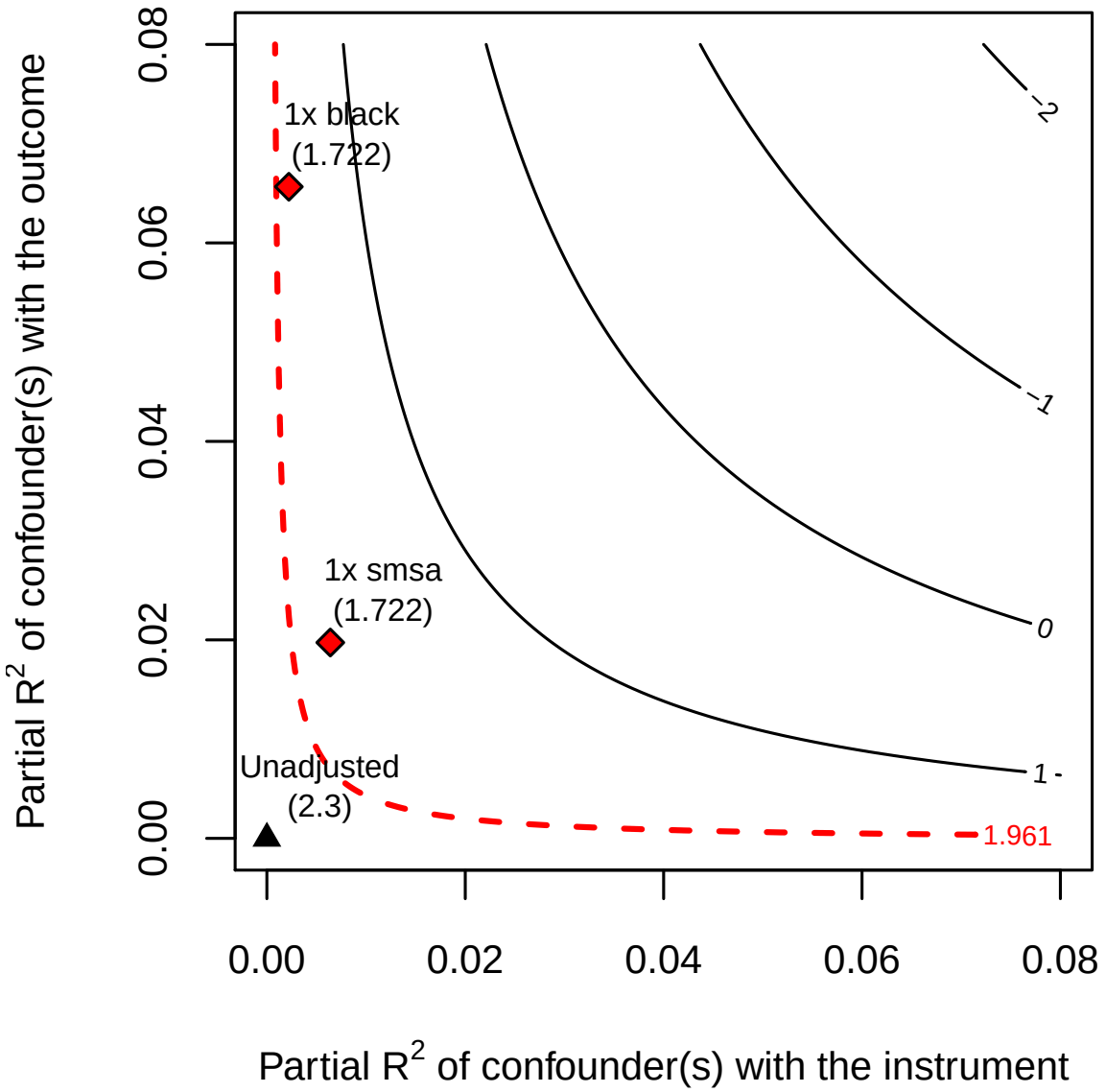
2. First stage: stability.

Outcome: *Education* (years)

Instrument	Est.	SE	t	$R^2_{D\sim Z\mid \boldsymbol{X}}$	$\text{XRV}_{q^*,\alpha}$	$\text{RV}_{q^*,\alpha}$
Proximity	0.32	0.088	3.64	0.44%	0.31%	3.02%
Bound (1× smsa): $R^2_{D\sim W\mid Z,\boldsymbol{X}} = 0.5\%$, $R^2_{Z\sim W\mid \boldsymbol{X}} = 0.6\%$, $t^\dagger = 2.26$						

Card example: reduced form contours



Card example: sensitivity of the IV (AR)

Outcome: *Earnings* (log)

Treatment	Est.	LL	UL	t	$\text{XRV}_{\geq q^*,\alpha}$	$\text{RV}_{\geq q^*,\alpha}$
Education	0.132	0.025	0.285	2.33	0.05%	0.67%

Bound (1× smsa): $R^2_{Y_0 \sim W|Z,\boldsymbol{X}} = 2\%$, $R^2_{W \sim Z|\boldsymbol{X}} = 0.6\%$, $t^\dagger = 2.55$

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- Identical to RF results because the null is $\tau = 0$.
- More interesting under other null hypotheses.

Card example: CI-limit contours

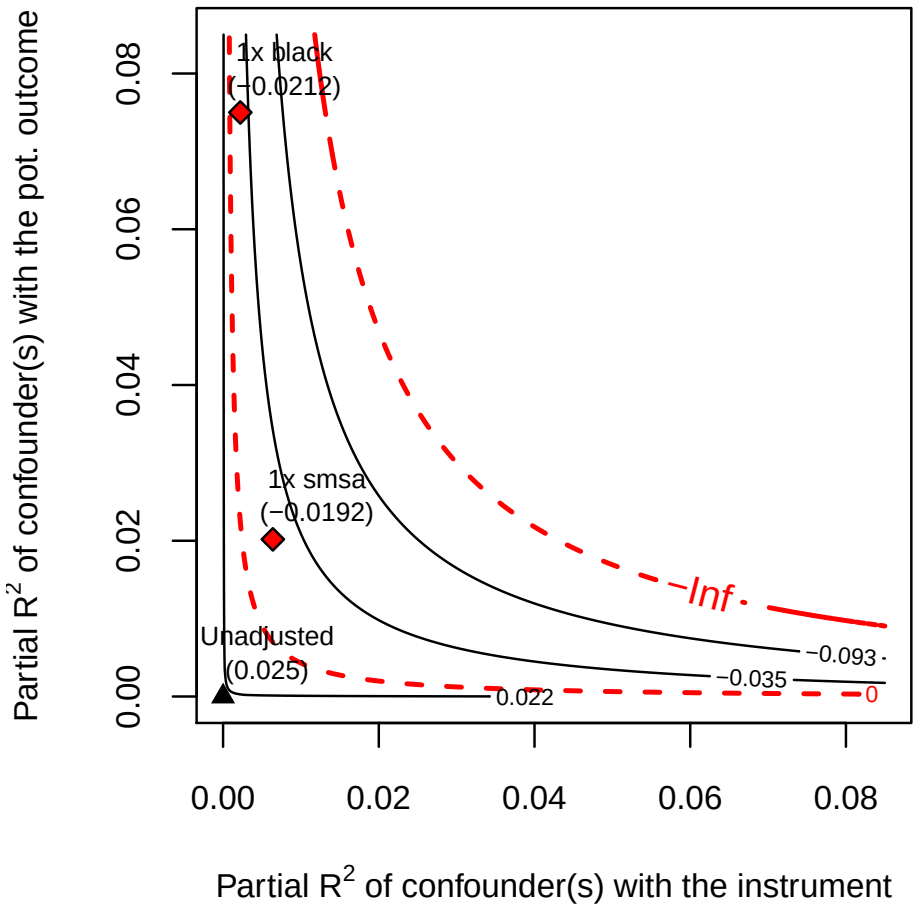

Figure: Sensitivity contours: lower 95% limit

IV sensitivity: conclusions

- OLS sensitivity to confounding can be as simple as adjusting your critical value.
- For null of zero in IV, sensitivity of the reduced form *is all you need*.
- General sensitivity for IV is also easy: in the AR framework, explore how postulated confounding moves point estimates, CI limits, RVs, and bounds.